

Brutal B49 - Electric Current, A Peek Beyond the Point (Decimals), Weather Climate & Adaptations, Symmetry

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A heater coil warms water at 4.5 degrees Celsius each minute when 2 cells drive it. With 3 cells the warming rate doubles. Starting at 26 C, what is the water's temperature after 4 minutes with 3 cells? [\[Electric Current & its Effects\]](#)
- (A) 36 C
 - (B) 53 C
 - (C) 62 C
 - (D) 26 C
 - (E) 47.5 C
 - (F) 44 C
 - (G) 98 C
2. A ribbon 2.5 metres long is cut into pieces each 0.35 metres long. How many full pieces can be cut, and what length of ribbon is left over? [\[A Peek Beyond the Point\]](#)
- (A) 6 pieces, 0.4 m left
 - (B) 7 pieces, 0.05 m left
 - (C) 7 pieces, 0.5 m left
 - (D) 5 pieces, 0.75 m left
 - (E) 7.14 pieces
 - (F) 7 pieces, 0.15 m left
3. The maximum temperatures from Monday to Friday were 34, 36, 33, 38 and 34 degrees Celsius. What is the mean maximum temperature for these 5 days? [\[Weather, Climate & Adaptations\]](#)
- (A) 36 C
 - (B) 35.5 C
 - (C) 33 C
 - (D) 37 C
 - (E) 35 C
 - (F) 34 C
 - (G) 38 C
4. Consider a regular hexagon. Find the number of its lines of symmetry, and add to it the order of its rotational symmetry. What is the total? [\[Symmetry\]](#)
- (A) 10
 - (B) 6
 - (C) 7
 - (D) 18
 - (E) 36
 - (F) 12
 - (G) 8

5. An electromagnet picks up 6 paper clips for every 50 turns of wire wound on its iron nail, in direct proportion. How many clips will 175 turns pick up? [\[Electric Current & its Effects\]](#)

- (A) 21 clips
- (B) 15 clips
- (C) 175 clips
- (D) 24 clips
- (E) 18 clips
- (F) 12 clips
- (G) 20 clips

6. When 0.6 of a certain number is taken, the result is 0.27. What is the number? [\[A Peek Beyond the Point\]](#)

- (A) 0.045
- (B) 0.9
- (C) 0.162
- (D) 0.45
- (E) 1.62
- (F) 4.5
- (G) 0.27

7. On one day the maximum was 39 C and the minimum 27 C; the day's range is maximum minus minimum. The next day the range is 3 C greater, with the same minimum. What is the next day's maximum? [\[Weather, Climate & Adaptations\]](#)

- (A) 39 C
- (B) 41 C
- (C) 12 C
- (D) 15 C
- (E) 45 C
- (F) 42 C

8. A shape has rotational symmetry of order 5. What is the smallest angle through which it can be turned to look exactly the same? [\[Symmetry\]](#)

- (A) 36 degrees
- (B) 45 degrees
- (C) 90 degrees
- (D) 5 degrees
- (E) 60 degrees
- (F) 72 degrees
- (G) 144 degrees

9. A 5 ampere fuse melts if the current crosses 5 A. On one line, four devices each draw 0.8 A and two more each draw 1.1 A. What is the total current, and does the fuse blow? [\[Electric Current & its Effects\]](#)

- (A) 5.4 A, fuse blows
- (B) 1.9 A, holds
- (C) 5.4 A, fuse holds
- (D) 6.6 A, blows
- (E) 4.3 A, holds
- (F) 3.2 A, holds

- 10.** First add 3.4, 0.85 and 12.7. Then subtract 6.95 from that sum. What is the final value? [\[A Peek Beyond the Point\]](#)
- (A) 11.0
 - (B) 10
 - (C) 10.10
 - (D) 17.0
 - (E) 4.0
 - (F) 16.95
- 11.** A town received 120 mm of rain in June, 250 mm in July and 180 mm in August. What is the total rainfall for these months, in centimetres? [\[Weather, Climate & Adaptations\]](#)
- (A) 0.55 cm
 - (B) 55 cm
 - (C) 5.5 cm
 - (D) 183.3 cm
 - (E) 550 cm
 - (F) 55 mm
- 12.** A regular polygon looks the same after the smallest turn of 24 degrees. How many sides does it have? [\[Symmetry\]](#)
- (A) 24 sides
 - (B) 12 sides
 - (C) 18 sides
 - (D) 15 sides
 - (E) 10 sides
 - (F) 8 sides
- 13.** A bulb glows at 0.6 A when 2 cells push the current, and the current rises in direct proportion to the number of cells. What current flows with 5 cells? [\[Electric Current & its Effects\]](#)
- (A) 1.5 A
 - (B) 0.9 A
 - (C) 0.6 A
 - (D) 1.8 A
 - (E) 1.2 A
 - (F) 2.5 A
 - (G) 3.0 A
- 14.** A car uses 5.6 litres of fuel to travel 80 kilometres. At the same rate, how many litres does it need for 350 kilometres? [\[A Peek Beyond the Point\]](#)
- (A) 28 L
 - (B) 19.6 L
 - (C) 22.4 L
 - (D) 0.07 L
 - (E) 49 L
 - (F) 35 L
 - (G) 24.5 L

15. A region gets 1,800 mm of rain in a year, and two-thirds of it falls during the 4 monsoon months. What is the average rainfall in one monsoon month? [\[Weather, Climate & Adaptations\]](#)

- (A) 300 mm
- (B) 150 mm
- (C) 200 mm
- (D) 250 mm
- (E) 600 mm
- (F) 1200 mm
- (G) 450 mm

16. An equilateral triangle is turned by 240 degrees about its centre. Its smallest turn that maps it onto itself is 120 degrees. How many such 120-degree turns is 240 degrees, and does the triangle land on itself?

[\[Symmetry\]](#)

- (A) 240 turns, yes
- (B) 2 turns, yes it maps onto itself
- (C) 2 turns, only partly
- (D) 1.5 turns, yes
- (E) 1 turn, yes
- (F) 3 turns, yes

17. A coil uses 0.25 units of electricity every 6 minutes. If it runs for 1 hour 12 minutes, how many units does it use? [\[Electric Current & its Effects\]](#)

- (A) 0.3 units
- (B) 18 units
- (C) 1.2 units
- (D) 3 units
- (E) 2.5 units
- (F) 12 units
- (G) 3.6 units

18. Multiply 0.025 by 0.4, and then divide the answer by 0.05. What do you get? [\[A Peek Beyond the Point\]](#)

- (A) 0.02
- (B) 2
- (C) 0.1
- (D) 0.0002
- (E) 0.05
- (F) 0.5
- (G) 0.2

19. The humidity reading rose from 40% to 70% during the day. By what percent did the humidity reading increase, compared with its starting value? [\[Weather, Climate & Adaptations\]](#)

- (A) 30%
- (B) 30 percentage points
- (C) 175%
- (D) 42.9%
- (E) 70%
- (F) 57%
- (G) 75%

- 20.** Among the capital letters H, E, X, A, G, O and N, how many have a vertical line of symmetry (a mirror line standing upright)? [\[Symmetry\]](#)
- (A) 7
 - (B) 2
 - (C) 5
 - (D) 6
 - (E) 1
 - (F) 4
- 21.** Electricity costs 7.50 rupees per unit. A geyser uses 2.4 units every day. What is the cost of running it for all 30 days of June? [\[Electric Current & its Effects\]](#)
- (A) 180 rupees
 - (B) 54 rupees
 - (C) 225 rupees
 - (D) 72 rupees
 - (E) 600 rupees
 - (F) 450 rupees
 - (G) 540 rupees
- 22.** Gold is priced at 6,250.50 rupees per gram. What is the cost of a piece weighing exactly 0.8 grams? [\[A Peek Beyond the Point\]](#)
- (A) 6250.40 rupees
 - (B) 781.31 rupees
 - (C) 5000.40 rupees
 - (D) 50004 rupees
 - (E) 500.04 rupees
 - (F) 5000.04 rupees
 - (G) 4000.40 rupees
- 23.** A polar bear's thick fur and fat suit a cold home where winter is minus 18 C and summer is 7 C. What is the difference between the summer and winter temperatures? [\[Weather, Climate & Adaptations\]](#)
- (A) 7 C
 - (B) minus 11 C
 - (C) 25 C
 - (D) minus 25 C
 - (E) 11 C
 - (F) 18 C
 - (G) 32 C
- 24.** Take a regular octagon. Add together its number of sides, its number of lines of symmetry, and its order of rotational symmetry. What is the sum? [\[Symmetry\]](#)
- (A) 18
 - (B) 32
 - (C) 20
 - (D) 16
 - (E) 24
 - (F) 48
 - (G) 8

- 25.** Doubling the current through an electromagnet makes it lift 3 times as much. At 0.4 A it lifts 150 grams. How much will it lift at 0.8 A? [\[Electric Current & its Effects\]](#)
- (A) 450 g
 - (B) 75 g
 - (C) 600 g
 - (D) 300 g
 - (E) 150 g
 - (F) 225 g
 - (G) 900 g
- 26.** A 1.5 litre bottle of juice is poured fully into small cups, each holding 0.125 litres. How many cups are filled? [\[A Peek Beyond the Point\]](#)
- (A) 12.5 cups
 - (B) 12 cups
 - (C) 1.2 cups
 - (D) 8 cups
 - (E) 6 cups
 - (F) 120 cups
 - (G) 15 cups
- 27.** A weather station takes one temperature reading every 3 hours over a 24-hour day, starting at midnight and stopping just before the next midnight. How many readings does it record? [\[Weather, Climate & Adaptations\]](#)
- (A) 3 readings
 - (B) 6 readings
 - (C) 24 readings
 - (D) 9 readings
 - (E) 7 readings
 - (F) 12 readings
 - (G) 8 readings
- 28.** A square has lines of symmetry along both diagonals and both mid-lines. A rectangle that is not a square has fewer. How many fewer lines of symmetry does the rectangle have than the square? [\[Symmetry\]](#)
- (A) 5
 - (B) 1
 - (C) 3
 - (D) 0
 - (E) 2
 - (F) 6
- 29.** An LED uses 9 watts and an old bulb uses 60 watts for the same brightness. If each runs for 5 hours, how many watt-hours does the LED save? [\[Electric Current & its Effects\]](#)
- (A) 69 Wh
 - (B) 45 Wh
 - (C) 51 Wh
 - (D) 270 Wh
 - (E) 300 Wh
 - (F) 255 Wh
 - (G) 345 Wh

30. Write the fraction $\frac{7}{8}$ as a decimal, then add 0.045 to it. What is the result? [\[A Peek Beyond the Point\]](#)

- (A) 0.92
- (B) 0.092
- (C) 0.9
- (D) 0.0795
- (E) 1.325
- (F) 0.42
- (G) 0.875

31. Over three months the average rainfall was 86 mm. Two of the months had 78 mm and 90 mm of rain. How much rain fell in the third month? [\[Weather, Climate & Adaptations\]](#)

- (A) 168 mm
- (B) 94 mm
- (C) 86 mm
- (D) 82 mm
- (E) 88 mm
- (F) 90 mm
- (G) 84 mm

32. A windmill with 4 identical blades has rotational symmetry of order 4 but no line of symmetry. What is the smallest angle through which it can be turned to look the same? [\[Symmetry\]](#)

- (A) 180 degrees
- (B) 90 degrees
- (C) 4 degrees
- (D) 120 degrees
- (E) 72 degrees
- (F) 360 degrees
- (G) 45 degrees

33. Bulbs joined one after another (in series) get dimmer. One bulb draws 0.45 A. Two such bulbs in series halve the current, three give one-third, and so on. What current flows with 3 bulbs in series? [\[Electric Current & its Effects\]](#)

- (A) 0.15 A
- (B) 1.35 A
- (C) 0.135 A
- (D) 0.9 A
- (E) 0.45 A
- (F) 0.225 A

34. Round 3.14159 to two decimal places, and then multiply the rounded value by 100. What do you get? [\[A Peek Beyond the Point\]](#)

- (A) 31415
- (B) 3.14
- (C) 314
- (D) 31.4
- (E) 314.159
- (F) 315
- (G) 300

35. A camel can drink 120 litres of water in 13 minutes at a steady rate. How many litres does it drink per minute, to one decimal place? [\[Weather, Climate & Adaptations\]](#)

- (A) 8.0 L/min
- (B) 9.2 L/min
- (C) 11.0 L/min
- (D) 9.0 L/min
- (E) 13 L/min
- (F) 0.11 L/min
- (G) 10.4 L/min

36. A clock's hour hand moves from the 12 to the 4. Through what angle has it turned, and what fraction of a full turn is that? [\[Symmetry\]](#)

- (A) 120 degrees, $\frac{1}{3}$ of a turn
- (B) 90 degrees, $\frac{1}{4}$ of a turn
- (C) 60 degrees, $\frac{1}{6}$ of a turn
- (D) 30 degrees, $\frac{1}{12}$ of a turn
- (E) 120 degrees, $\frac{1}{2}$ of a turn
- (F) 120 degrees, $\frac{1}{4}$ of a turn
- (G) 240 degrees, $\frac{2}{3}$ of a turn

37. An electric bell strikes 25 times in 10 seconds. At the same steady rate, how many times does it strike in 2.5 minutes? [\[Electric Current & its Effects\]](#)

- (A) 37 strikes
- (B) 625 strikes
- (C) 375 strikes
- (D) 60 strikes
- (E) 150 strikes
- (F) 250 strikes

38. A digital thermometer reads 36.6 degrees Celsius. The reading rises by 1.8, then falls by 0.45. What is the final reading? [\[A Peek Beyond the Point\]](#)

- (A) 39.0 C
- (B) 37.5 C
- (C) 37.05 C
- (D) 38.4 C
- (E) 37.95 C
- (F) 36.95 C
- (G) 38.85 C

39. Near spring, the length of daylight grows by 2 minutes 15 seconds each day. By how much does daylight grow over 8 days, in minutes and seconds? [\[Weather, Climate & Adaptations\]](#)

- (A) 18 min 15 s
- (B) 18 min 0 s
- (C) 16 min 15 s
- (D) 16 min 0 s
- (E) 18 min 2 s
- (F) 20 min 0 s
- (G) 17 min 0 s

40. A regular pentagon is turned three times, each time by its smallest angle of rotational symmetry. Through what total angle has it turned? [\[Symmetry\]](#)

- (A) 180 degrees
- (B) 72 degrees
- (C) 108 degrees
- (D) 216 degrees
- (E) 288 degrees
- (F) 360 degrees
- (G) 144 degrees

41. An LED draws 20 milliamperes of current. A decorative strip has 45 such LEDs sharing the supply equally per LED. What total current, in amperes, does the strip draw? [\[Electric Current & its Effects\]](#)

- (A) 90 A
- (B) 0.09 A
- (C) 0.9 A
- (D) 1.8 A
- (E) 9 A
- (F) 900 A
- (G) 0.45 A

42. In the number 408.073, what value does the digit 7 stand for? [\[A Peek Beyond the Point\]](#)

- (A) 7 thousandths (0.007)
- (B) 0.7
- (C) 0.07 (seven hundredths)
- (D) 7 tenths (0.7)
- (E) 0.0007
- (F) 70

43. Air temperature falls by 0.65 C for every 100 metres a climber goes up. Starting at 30 C, what is the temperature after climbing 1,500 metres? [\[Weather, Climate & Adaptations\]](#)

- (A) 9.75 C
- (B) 20.25 C
- (C) 19.75 C
- (D) 20.75 C
- (E) 29.35 C
- (F) 18.75 C
- (G) 21.75 C

44. Multiply the number of lines of symmetry of a regular 12-sided polygon by the order of rotational symmetry of a regular (equilateral) triangle. What is the product? [\[Symmetry\]](#)

- (A) 9
- (B) 24
- (C) 12
- (D) 4
- (E) 36
- (F) 48

45. A device draws 6.4 A. The shop sells fuses rated 5 A, 8 A, 10 A and 13 A. The safe fuse must be the smallest rating that is still above the device current. Which fuse should be chosen? [\[Electric Current & its Effects\]](#)

- (A) 5 A
- (B) 8 A
- (C) 7 A
- (D) 6.4 A
- (E) 13 A
- (F) 6 A
- (G) 10 A

46. Coins, each 0.8 millimetres thick, are stacked into a pile 3.2 centimetres tall. How many coins are in the pile? [\[A Peek Beyond the Point\]](#)

- (A) 400 coins
- (B) 4 coins
- (C) 25 coins
- (D) 40 coins
- (E) 16 coins
- (F) 26 coins
- (G) 32 coins

47. In a desert the daytime temperature is 48 C and the night temperature is 12 C. The night temperature is what fraction of the day temperature, in lowest terms? [\[Weather, Climate & Adaptations\]](#)

- (A) $\frac{1}{3}$
- (B) $\frac{3}{4}$
- (C) $\frac{2}{5}$
- (D) $\frac{1}{2}$
- (E) 4
- (F) $\frac{1}{4}$
- (G) $\frac{1}{5}$

48. A figure looks exactly the same after a one-sixth turn. After how many such equal turns does it first return to its start, and what is the smallest turning angle? [\[Symmetry\]](#)

- (A) 6 turns, 60 degrees
- (B) 3 turns, 120 degrees
- (C) 6 turns, 36 degrees
- (D) 12 turns, 30 degrees
- (E) 6 turns, 6 degrees
- (F) 1 turn, 360 degrees

49. A kettle raises 2 litres of water by 70 degrees Celsius using 0.16 units of electricity. Using the same kettle, how many units are needed to heat 5 litres by the same 70 C? [\[Electric Current & its Effects\]](#)

- (A) 0.8 units
- (B) 0.5 units
- (C) 1.0 units
- (D) 0.064 units
- (E) 0.4 units
- (F) 0.32 units
- (G) 0.16 units

50. Subtract 0.7 from 5, and then divide the result by 0.25. What number do you get? [\[A Peek Beyond the Point\]](#)

- (A) 4.3
- (B) 17.2
- (C) 1.075
- (D) 1.72
- (E) 16
- (F) 18
- (G) 43

51. Snow falls at 4.5 centimetres per hour for 3.5 hours, and then 6 centimetres of it melts. What depth of snow is left? [\[Weather, Climate & Adaptations\]](#)

- (A) 3.75 cm
- (B) 21.75 cm
- (C) 10.5 cm
- (D) 9.5 cm
- (E) 9.75 cm
- (F) 8.25 cm
- (G) 15.75 cm

52. Half of a butterfly is drawn with a vertical line of symmetry down the middle. If the left wing tip is 5.4 centimetres from the line, how far apart are the two wing tips? [\[Symmetry\]](#)

- (A) 10.4 cm
- (B) 8.1 cm
- (C) 21.6 cm
- (D) 2.7 cm
- (E) 5.4 cm
- (F) 10.8 cm
- (G) 5.04 cm

53. A ceiling fan is rated to use 0.075 units of electricity per hour. If it runs 8 hours every day, how many units does it use in a 31-day month? [\[Electric Current & its Effects\]](#)

- (A) 0.6 units
- (B) 18.0 units
- (C) 24.8 units
- (D) 186 units
- (E) 2.4 units
- (F) 18.6 units

54. A recipe needs 0.375 kilograms of flour for one cake. From a 2-kilogram bag, how many whole cakes can be made, and how many grams of flour are left? [\[A Peek Beyond the Point\]](#)

- (A) 6 cakes, 0 g left
- (B) 5 cakes, 125 g left
- (C) 5 cakes, 12.5 g left
- (D) 5 cakes, 250 g left
- (E) 5.33 cakes
- (F) 4 cakes, 500 g left
- (G) 5 cakes, 1.25 g left

- 55.** Migratory birds fly 6,400 kilometres in 8 days, covering an equal distance each day. How far do they travel in 5 days? [\[Weather, Climate & Adaptations\]](#)
- (A) 800 km
 - (B) 3,200 km
 - (C) 1,280 km
 - (D) 2,000 km
 - (E) 4,000 km
 - (F) 5,120 km
 - (G) 4,500 km
- 56.** An equilateral triangle has a certain number of lines of symmetry, and a scalene triangle has a certain number. Multiply these two counts together. What is the product? [\[Symmetry\]](#)
- (A) 0
 - (B) 1
 - (C) 6
 - (D) 2
 - (E) 9
 - (F) 12
 - (G) 3
- 57.** A kitchen circuit-breaker trips if the current goes above 16 A. An oven draws 9.5 A, a microwave 4.8 A and a toaster 3.2 A, all switched on together. By how much is the 16 A limit exceeded? [\[Electric Current & its Effects\]](#)
- (A) 16 A
 - (B) 1.4 A
 - (C) 17.5 A
 - (D) 2.5 A
 - (E) 0.5 A
 - (F) 3.5 A
 - (G) 1.5 A
- 58.** What is 0.04 multiplied by 0.04? [\[A Peek Beyond the Point\]](#)
- (A) 0.00016
 - (B) 0.0016
 - (C) 0.016
 - (D) 1.6
 - (E) 0.16
 - (F) 0.0004
- 59.** On one day the highest temperature was 41 C and the lowest was 29 C. What is the average (mean) of the highest and lowest temperatures? [\[Weather, Climate & Adaptations\]](#)
- (A) 12 C
 - (B) 40 C
 - (C) 70 C
 - (D) 35 C
 - (E) 36 C
 - (F) 6 C

60. A regular polygon has the same number of lines of symmetry as it has sides. It has 10 lines of symmetry. What is the smallest angle through which it can be turned to look the same? [\[Symmetry\]](#)

- (A) 30 degrees
- (B) 36 degrees
- (C) 18 degrees
- (D) 10 degrees
- (E) 45 degrees
- (F) 144 degrees
- (G) 72 degrees

61. Working alone, heating coil A warms a cup of water in 4 minutes and coil B warms the same cup in 6 minutes. If both work together their warming rates add. How long do they take together? [\[Electric Current & its Effects\]](#)

- (A) 5 minutes
- (B) 10 minutes
- (C) 2 minutes
- (D) 2.4 minutes
- (E) 3 minutes
- (F) 0.5 minutes

62. Find the average (mean) of the three numbers 2.5, 3.75 and 4.25. [\[A Peek Beyond the Point\]](#)

- (A) 3.75
- (B) 3.6
- (C) 3.0
- (D) 3.5
- (E) 10.5
- (F) 10.50

63. Wind makes the air feel colder: the felt temperature drops by 1.5 C for every 10 km/h of wind. If the air is 8 C and the wind is 40 km/h, what is the felt temperature? [\[Weather, Climate & Adaptations\]](#)

- (A) 2 C
- (B) minus 2 C
- (C) 3.5 C
- (D) 14 C
- (E) 0 C
- (F) minus 4 C
- (G) 6 C

64. Of these four shapes, how many have at least one line of symmetry: a rectangle, a parallelogram that is not a rectangle, a rhombus, and a scalene triangle? [\[Symmetry\]](#)

- (A) 0
- (B) all four
- (C) 4
- (D) 2
- (E) 1
- (F) 2 and a half
- (G) 3

- 65.** Turning off appliances left on standby saves 0.8 units of electricity each day. The home used 6.4 units a day before. What percent of the daily electricity is saved? [\[Electric Current & its Effects\]](#)
- (A) 6.4%
 - (B) 16%
 - (C) 8%
 - (D) 20%
 - (E) 12.5%
 - (F) 0.8%
 - (G) 80%
- 66.** A plank is 4.2 metres long. Three pieces, each 0.95 metres long, are cut from it. What length of plank remains? [\[A Peek Beyond the Point\]](#)
- (A) 2.85 m
 - (B) 1.35 m
 - (C) 1.95 m
 - (D) 1.25 m
 - (E) 3.25 m
 - (F) 2.35 m
 - (G) 0.35 m
- 67.** In one year 35% of a place's total 2,000 mm of rain fell during July. How much rain fell in July, in centimetres? [\[Weather, Climate & Adaptations\]](#)
- (A) 35 cm
 - (B) 70 cm
 - (C) 7 cm
 - (D) 57 cm
 - (E) 700 mm
 - (F) 65 cm
- 68.** A square tile is rotated 270 degrees clockwise and looks exactly the same. Its smallest symmetry turn is 90 degrees. How many 90-degree steps is 270 degrees, and does the tile map onto itself? [\[Symmetry\]](#)
- (A) 3 steps, no
 - (B) 3 steps, only partly
 - (C) 270 steps, yes
 - (D) 3 steps, yes it maps onto itself
 - (E) 2 steps, yes
 - (F) 4 steps, yes
 - (G) 1 step, yes
- 69.** A bulb's coiled filament is 0.012 metres long when stretched out. A pack holds 8 identical bulbs. What is the total filament length, in centimetres? [\[Electric Current & its Effects\]](#)
- (A) 96 cm
 - (B) 12 cm
 - (C) 0.96 cm
 - (D) 0.096 cm
 - (E) 9.6 cm
 - (F) 0.0096 cm

70. Write the decimal 0.625 as a fraction in its lowest terms. [\[A Peek Beyond the Point\]](#)

- (A) $\frac{5}{8}$
- (B) $\frac{625}{1000}$
- (C) $\frac{2}{3}$
- (D) $\frac{3}{5}$
- (E) $\frac{1}{16}$
- (F) $\frac{5}{80}$

71. A frog hibernates as a pond cools. The water cools steadily at 1.2 C per hour. How many hours does it take to cool from 19 C down to 4 C? [\[Weather, Climate & Adaptations\]](#)

- (A) 12.5 hours
- (B) 10 hours
- (C) 15.8 hours
- (D) 11 hours
- (E) 12 hours
- (F) 18 hours
- (G) 13.5 hours

72. For a rectangle that is not a square, add its order of rotational symmetry to its number of lines of symmetry. What is the total? [\[Symmetry\]](#)

- (A) 0
- (B) 2
- (C) 6
- (D) 1
- (E) 8
- (F) 4
- (G) 3

73. A solar lamp runs for 8 hours after a full 5-hour charge, and run time stays in proportion to charge time. On a cloudy day it charges for only 3.5 hours. For how long can it then run? [\[Electric Current & its Effects\]](#)

- (A) 7 h
- (B) 5 h 36 min
- (C) 8 h
- (D) 4 h 24 min
- (E) 5 h 6 min
- (F) 5 h 30 min

74. A runner covers 100 metres in 12.5 seconds. What is the runner's speed in metres per second, to one decimal place? [\[A Peek Beyond the Point\]](#)

- (A) 0.125 m/s
- (B) 12.5 m/s
- (C) 7.5 m/s
- (D) 8.0 m/s
- (E) 8.3 m/s
- (F) 6.0 m/s

75. City A has an average temperature of 28 C and City B averages 22 C. City A is what percent warmer than City B, to one decimal place? [\[Weather, Climate & Adaptations\]](#)

- (A) 75%
- (B) 27.3%
- (C) 6%
- (D) 6.0%
- (E) 21.4%
- (F) 127%

76. A regular star with 8 equal points has rotational symmetry of order 8. Through what total angle is it turned by 5 of its smallest symmetry turns? [\[Symmetry\]](#)

- (A) 320 degrees
- (B) 225 degrees
- (C) 40 degrees
- (D) 45 degrees
- (E) 360 degrees
- (F) 200 degrees

77. Four identical cells joined in series give 6 volts in total. A toy needs 9 volts to work. How many such cells in series are needed? [\[Electric Current & its Effects\]](#)

- (A) 3 cells
- (B) 12 cells
- (C) 4 cells
- (D) 9 cells
- (E) 5 cells
- (F) 6 cells
- (G) 8 cells

78. With a single 50-rupee note you buy items costing 12.75 rupees each. What is the greatest number you can buy, and how much change is left? [\[A Peek Beyond the Point\]](#)

- (A) 3 items, 12.75 rupees change
- (B) 4 items, 0 change
- (C) 3 items, 11.75 rupees change
- (D) 4 items, 1 rupee change
- (E) 2 items, 24.50 rupees change
- (F) 3 items, 11.25 rupees change
- (G) 3.92 items

79. Four humidity readings 60%, 65%, 70% and one more have a mean of 63%. What is the missing fourth reading? [\[Weather, Climate & Adaptations\]](#)

- (A) 70%
- (B) 57%
- (C) 55%
- (D) 62%
- (E) 68%
- (F) 65%
- (G) 63%

- 80.** A regular hexagon and a square are placed side by side. Counting each shape on its own, what is the total number of lines of symmetry of the two shapes together? [\[Symmetry\]](#)
- (A) 6
 - (B) 14
 - (C) 10
 - (D) 4
 - (E) 12
 - (F) 8
- 81.** In a heating wire, the heat made each second grows with the square of the current. At 2 A it makes 8 joules each second. How much heat each second does it make at 3 A? [\[Electric Current & its Effects\]](#)
- (A) 16 J/s
 - (B) 18 J/s
 - (C) 24 J/s
 - (D) 8 J/s
 - (E) 12 J/s
 - (F) 36 J/s
 - (G) 6 J/s
- 82.** Add the decimals $0.3 + 0.03 + 0.003 + 0.0003$. What is the sum? [\[A Peek Beyond the Point\]](#)
- (A) 0.012
 - (B) 0.333
 - (C) 0.33
 - (D) 0.34
 - (E) 3.333
 - (F) 0.3333
- 83.** A glacier retreats by 0.025 kilometres every year. How many metres does it retreat in 8 years? [\[Weather, Climate & Adaptations\]](#)
- (A) 2 m
 - (B) 100 m
 - (C) 0.2 m
 - (D) 200 m
 - (E) 2,000 m
 - (F) 20 m
 - (G) 0.02 m
- 84.** A shape maps onto itself when turned by 40 degrees. What is its order of rotational symmetry? [\[Symmetry\]](#)
- (A) 9
 - (B) 8
 - (C) 6
 - (D) 10
 - (E) 40
 - (F) 320
 - (G) 18

85. Resistance wire is sold by weight: 2.5 metres of it weighs 30 grams. A heater needs 7 coils, each 0.8 metres long. What is the total mass of wire used? [\[Electric Current & its Effects\]](#)

- (A) 84 g
- (B) 168 g
- (C) 13.44 g
- (D) 30 g
- (E) 60 g
- (F) 67.2 g
- (G) 56 g

86. A melon weighs 2.4 kilograms and is priced at 3.50 rupees per 100 grams. What is its total cost? [\[A Peek Beyond the Point\]](#)

- (A) 42 rupees
- (B) 84 rupees
- (C) 840 rupees
- (D) 60 rupees
- (E) 72 rupees
- (F) 8.40 rupees
- (G) 8.75 rupees

87. On a certain day there are 14 hours of daylight and the rest is night. What is the ratio of daylight to night, in its simplest form? [\[Weather, Climate & Adaptations\]](#)

- (A) 14:10
- (B) 10:14
- (C) 12:12
- (D) 7:5
- (E) 3:2
- (F) 14:24
- (G) 5:7

88. A regular pentagon has a number of lines of symmetry, and a regular hexagon has a number of lines of symmetry. How many more lines of symmetry does the hexagon have than the pentagon? [\[Symmetry\]](#)

- (A) 1
- (B) 2
- (C) 11
- (D) 0
- (E) 6
- (F) 5

89. Bulbs come in packs of 3 for 48 rupees, and packs are sold only whole. A school must replace 17 bulbs. What is the least amount it can spend? [\[Electric Current & its Effects\]](#)

- (A) 240 rupees
- (B) 304 rupees
- (C) 256 rupees
- (D) 288 rupees
- (E) 816 rupees
- (F) 272 rupees
- (G) 96 rupees

90. Which of these decimals is the largest: 0.4, 0.39, 0.409 or 0.4001? [\[A Peek Beyond the Point\]](#)

- (A) 0.409
- (B) they are all equal
- (C) 0.4
- (D) 0.39 and 0.4 tie
- (E) 0.4001
- (F) 0.04

91. A region's yearly rainfall doubled between 2018 and 2023, an increase of 640 mm. What was the rainfall in 2023? [\[Weather, Climate & Adaptations\]](#)

- (A) 1,000 mm
- (B) 640 mm
- (C) 960 mm
- (D) 1,280 mm
- (E) 1,920 mm
- (F) 2,560 mm
- (G) 320 mm

92. An isosceles triangle has exactly one line of symmetry. This line of symmetry passes through the top vertex and which angle does it cut into two equal halves? [\[Symmetry\]](#)

- (A) the largest side
- (B) the apex (top) angle
- (C) all three angles
- (D) no angle at all
- (E) a base angle
- (F) the right angle
- (G) both base angles

93. An immersion rod warms a bucket of water by 8 degrees Celsius every 3 minutes. The water must rise from 22 C to 50 C. How long, in minutes and seconds, does this take? [\[Electric Current & its Effects\]](#)

- (A) 14 min 0 s
- (B) 12 min 0 s
- (C) 9 min 0 s
- (D) 10 min 5 s
- (E) 10 min 30 s
- (F) 21 min 0 s

94. A tank leaks at a steady 0.015 litres every minute. How many litres leak out in one full day (24 hours)?

[\[A Peek Beyond the Point\]](#)

- (A) 21.6 L
- (B) 0.9 L
- (C) 2.16 L
- (D) 1.44 L
- (E) 0.36 L
- (F) 216 L
- (G) 36 L

- 95.** A barrel cactus holds 18 litres of stored water and loses 0.6% of it in one day. How many litres does it lose in that day? [\[Weather, Climate & Adaptations\]](#)
- (A) 1.08 L
 - (B) 0.108 L
 - (C) 10.8 L
 - (D) 1.8 L
 - (E) 0.6 L
 - (F) 0.18 L
 - (G) 0.0108 L
- 96.** A regular 20-sided figure can be turned by its smallest symmetry angle. What is that angle, and how many such turns make a half-circle of 180 degrees? [\[Symmetry\]](#)
- (A) 36 degrees, 5 turns
 - (B) 18 degrees, 10 turns
 - (C) 18 degrees, 20 turns
 - (D) 9 degrees, 20 turns
 - (E) 20 degrees, 9 turns
 - (F) 18 degrees, 18 turns
 - (G) 18 degrees, 5 turns
- 97.** A power bank gains 1.5 units of stored energy each hour while charging, but slowly leaks 0.3 units each hour at the same time. Starting empty, how many hours does it take to store 18 units? [\[Electric Current & its Effects\]](#)
- (A) 20 hours
 - (B) 9 hours
 - (C) 18 hours
 - (D) 15 hours
 - (E) 12 hours
 - (F) 10 hours
- 98.** Find half of 0.05, and add it to a quarter of 0.2. What is the total? [\[A Peek Beyond the Point\]](#)
- (A) 0.25
 - (B) 0.75
 - (C) 0.0075
 - (D) 0.105
 - (E) 0.075
 - (F) 0.0125
- 99.** The mean temperature over a week is 31 C. If every single day had been 2 C hotter, what would the new mean temperature be? [\[Weather, Climate & Adaptations\]](#)
- (A) 62 C
 - (B) 33.5 C
 - (C) 45 C
 - (D) 33 C
 - (E) 32 C
 - (F) 29 C
 - (G) 31 C

100. A regular hexagon has rotational symmetry of order 6, so its smallest turn is 60 degrees. If it is rotated by 420 degrees, the result is the same as which smaller single turn? [\[Symmetry\]](#)

- (A) back to the exact start (0 degrees)
- (B) the same as a 60-degree turn
- (C) the same as a 180-degree turn
- (D) the same as a 240-degree turn
- (E) the same as a 420-degree turn
- (F) the same as a 90-degree turn